

Lab: Synaptic Plasticity as Model Updating

Learning Goal: Map neural learning mechanisms to Active Inference parameter and structure learning.

Part 1: Synaptic Plasticity Rule Mapping

Exercise: Map each plasticity rule to the Active Inference operation it implements:

Plasticity Rule	Neural Mechanism	Active Inference Operation	Matrix Updated	Example
LTP (Hebbian)	Co-activation strengthens synapse	Increase association parameter	A matrix (observation likelihood)	"Seeing red co-occurs with seeing an apple" → strengthen red-apple link
LTD	Failed co-activation weakens synapse	Decrease association parameter	A matrix	"Expected green but saw red" → weaken green-apple link
STDP (causal LTP)	Pre-before-post → strengthen	Learn causal transition	B matrix (state transitions)	"Pressing button causes light" → strengthen button→light transition
STDP (acausal LTD)	Post-before-pre → weaken	Unlearn spurious causation	B matrix	"Light doesn't cause button press" → weaken light→button link
Dopaminergic RPE	Phasic DA encodes reward PE	Update value/preference	C vector (preferences)	"This food was tastier than expected" → increase food-preference
Cerebellar LTD	Climbing fiber PE at PC synapse	Update forward model	Forward model parameters	"My reach was off by 2cm" → adjust reach prediction

Write a 200-word analysis: How do these different plasticity rules, taken together, enable a complete generative model to be learned from scratch?

Write your response here

Part 2: Memory Consolidation Timeline

Learning Goal: Trace the consolidation process from encoding to long-term storage.

Exercise: Trace a specific memory (learning someone's name at a party) through the consolidation timeline:

Time Point	Brain System	Process	What Is Happening to the Generative Model
T=0 (encoding)	Hippocampus (CA3/CA1)	Rapid synaptic change	New parameter: this face → this name (fast, fragile)
T=30min	Hippocampus	Initial consolidation	Protein synthesis stabilizes new synapses
T=8hr (first sleep)	Hippocampus → Neocortex	SWR replay + spindle-gated LTP	Parameter begins transfer to cortical face model
Days 1-7	Gradual transfer	Repeated replay across multiple sleep cycles	Cortical parameters strengthen; hippocampal trace weakens
Weeks-months	Neocortex	Semantic integration	Name becomes part of general social knowledge
Reactivation	Hippocampus (reconsolidation)	Memory becomes labile	Window opens for parameter revision

What happens at each stage if the process is disrupted (e.g., sleep deprivation at T=8hr)?

Write your response here

Part 3: Sleep as Bayesian Model Reduction

Learning Goal: Analyze how sleep optimizes the generative model.

Exercise: Consider a student who studies intensely for an exam. Compare their brain state before and after sleep:

Before sleep:

- Many new synaptic connections formed during study (high parameter count)
- Some connections encode genuine patterns; others encode noise or irrelevant details
- Model is complex and potentially overfitting

During sleep:

1. SWR replay selectively reactivates the most important patterns
2. Global synaptic downscaling (synaptic homeostasis hypothesis) weakens all connections proportionally
3. Connections encoding strong, reliable patterns survive the downscaling
4. Connections encoding noise fall below threshold and are effectively pruned

After sleep:

- Fewer parameters but better generalization
- The student can answer novel questions (not just memorized ones)

Write a 300-word analysis: How does this process implement Bayesian Model Reduction? How does it relate to Occam's Razor? Why is sleep deprivation harmful to learning?

Write your response here

Part 4: Structure Learning Analysis

Learning Goal: Analyze how the brain adds new representational capacity.

Exercise: Consider a person who moves to a new city. How does the hippocampus create a new "map" (generative model) for the new environment?

1. **Neurogenesis:** New neurons born in dentate gyrus — what purpose do they serve?
2. **Place field formation:** How quickly do new place cells form? What drives them?
3. **Pattern separation:** How does the new map avoid interfering with the old city's map?
4. **Integration:** How does the new map get integrated into the broader generative model (e.g., knowing both cities and how to navigate between them)?

Write your response here

Part 5: Experimental Design

Learning Goal: Design an experiment testing Active Inference learning predictions.

Exercise: Design an experiment to test whether sleep-dependent synaptic downscaling implements BMR:

1. **Learning task:** What task would participants learn? (e.g., statistical learning, pattern recognition)
2. **Measurement:** How would you measure generalization vs. memorization?
3. **Sleep manipulation:** How would you manipulate sleep? (full sleep, sleep deprivation, targeted memory reactivation)
4. **Neural measurement:** What brain signals would you record? (spindle density, SWR rate, synaptic markers)
5. **Predictions:** What specific result would support the BMR hypothesis?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Mechanistic mapping	Plasticity rules → model updates
2	Timeline tracing	Memory consolidation process
3	Theoretical analysis	Sleep as BMR
4	Systems reasoning	Structure learning
5	Experiment design	Testing the BMR hypothesis